

Lemma. Let V be a fin. dim v.s over F , and let $T: V \rightarrow V$ be a linear operator. Suppose that the minimal polynomial of T splits over F as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper T -invariant subspace $W < V$, there exists a vector $\alpha \in V$ s.t. $\alpha \notin W$ and for some c_i , $(T - c_i I)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given, and let $g \in F[x]$ be a T -conductor of β into W . Then, g divides p . And, since $\beta \notin W$, g is not constant. Therefore

$$g = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k} \quad (0 \leq e_i \leq r_i)$$

and at least one of e_i is positive. Pick i s.t. $e_i > 0$. Then,

$$g = (x - c_i) \cdot h \quad \text{where } h = g / (x - c_i) \in F[x].$$

Since g is T -conductor of β , $h(T)(\beta) \notin W$, but $(T - c_i I)(h(T)(\beta)) = g(T)(\beta) \in W$.

Thm. Let V be a finite dimensional vector space over F , and $T: V \rightarrow V$ be a linear. T is Triangularizable if and only if minimal polynomial of T splits over F .

Proof. If T is triangularizable with respects to a basis β , then the characteristic polynomial is

$$f(x) = \det([T]_{\beta} - xI) = \prod_{i=1}^n (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{th diagonal element of } [T]_{\beta}$$

thus, proved. Conversely, choose $\alpha_1 \in V$ s.t. $\alpha_1 \notin \{0\}$, $(T - c_1 I)(\alpha_1) \in \{0\}$ for some eigenvalue $c_1 \in F$ of T . Choose $\alpha_2 \in V$ s.t. $\alpha_2 \notin \langle \alpha_1 \rangle$, $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, it is possible because $\langle \alpha_1 \rangle$ is T -invariant. Again, choose $\alpha_3 \in V$ s.t. $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$, $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$. It is possible again, $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$.

Now, we obtain $\{\alpha_1, \dots, \alpha_n\}$ s.t.

$$T(\alpha_1) = c_1 \alpha_1$$

$$T(\alpha_2) = a_{21} \alpha_1 + c_2 \alpha_2$$

$\vdots$

$$T(\alpha_n) = a_{n1} \alpha_1 + a_{n2} \alpha_2 + \dots + c_n \alpha_n$$

Thm: Let V be a fin. d'im. v.s over F , and $T: V \rightarrow V$ be a linear operator.
 T is diagonalizable if and only if minimal polynomial of T splits and separable.

Proof. Suppose that T is diagonalizable. Let $c_1, \dots, c_k$ be distinct eigenvalues.
 Then, for any eigenvector $d \in V$, there is a $c_i \in F$ s.t. $(T - c_i I)(d) = 0$.
 Therefore, for $p = (x - c_1) \dots (x - c_k)$,

$$p(T) = (T - c_1 I) \dots (T - c_k I)(d) = 0$$

because $\{T - c_1 I, \dots, T - c_k I\}$ commute. Since T is diagonalizable,
 for any vector of V can be represented by linear combination of eigenvectors,
 $p(T) = 0$

Thus proved. Conversely, let W be a spanned subspace by all of eigenvectors of T .
 Suppose that $W \neq V$. By the construction, W is T -invariant. Therefore, by the Lemma,
 there is $\alpha \in V \setminus W$ and eigenvalue $c_0 \in F$ s.t. $\beta = (T - c_0 I)(\alpha) \in W$.
 Since $\beta \in W$, $\beta = \beta_1 + \dots + \beta_k$ where $T(\beta_i) = c_i \beta_i$, $1 \leq i \leq k$. Therefore, for any $h \in F[x]$

$$h(T)(\beta) = \sum_{i=1}^k h(c_i) \cdot \beta_i \in W.$$

Put $p = (x - c_0)g$. Also, $g - g(c_0) = (x - c_0)h$. Then,

$$g(T)(\alpha) - g(c_0)(\alpha) = h(T)(T - c_0 I)(\alpha) = h(T)(\beta) \in W$$

And, $0 = p(T)(\alpha) = (T - c_0 I)g(T)(\alpha) = g(T)(T - c_0 I)(\alpha) \in W$.

Therefore $g(c_0)(\alpha) \in W$, and since $\alpha \notin W$, $g(c_0) = 0$. Contradicts to $p = (x - c_0)^2 g$.